$$\int d^4\widetilde{x}\ \sqrt{-\widetilde{g}(\widetilde{x})}\left[A(\widetilde{t}-\xi^0(x(\widetilde{x})))+B(\widetilde{t}-\xi^0(x(\widetilde{x})))\frac{\partial(\widetilde{t}-\xi^0(x(\widetilde{x})))}{\partial\widetilde{x}^\mu}\frac{\partial(\widetilde{t}-\xi^0(x(\widetilde{x})))}{\partial\widetilde{x}^\nu}\widetilde{g}^{\mu\nu}(\widetilde{x})\right] \quad (177)$$